

Table 2: Key Open Source Blockchain Platforms

Characteristic	Open source product	Features
Public blockchain	Ethereum, Ripple, Litecoin, etc	Permissionless blockchain; participants are anonymous; requires mining
Private blockchain	Hyperledger projects, Exo-num, OpenChain, MultiChain	Permissioned blockchain; identity of the participants is known before; no need for mining
Federated blockchain (Consortium)	R3, EWF, etc	Group of companies or individuals coming together to make decisions for the best benefit of the whole network; such groups are also called consortiums or a federated blockchain
Smart contracts in general programming languages	Hyperledger, Ethereum	Ethereum does not directly support Java; it does so with the help of a Web3j framework
No mining and native cryptocurrency	Hyperledger, Corda, Open-chain	Typically, public blockchains require mining (as part of proof of work) to add a block to the chain; miners are rewarded using native cryptocurrency
Pluggable consensus algorithms	Hyperledger, Corda, Open-chain	Support for pluggable consensus used for different applications
Pluggable ordering service	Hyperledger	Support for pluggable ordering service used for different applications.
Pluggable membership service	Hyperledger	Support for pluggable membership used for onboarding new users onto the network

commands on a blockchain if needed.

Hybrid/federated blockchain: This blockchain should be able to connect the public blockchain, which is open to every single person in the world, with a private blockchain, which runs in a fully permissioned environment, thus limiting the access to available information.

Conducting business over a decentralised hybrid blockchain reduces transaction costs, eliminates data redundancy, and speeds up transaction times.

Benefits and drawbacks of blockchain

- Transaction histories are becoming more transparent using blockchain technology
- Blockchain is more secure than other record-keeping systems
- Elimination of error handling through real-time tracking of transactions with no double spending
- Improved traceability
- Trusted record-keeping
- Increased efficiency and speed, reduction of settlement time to mere seconds by removing intermediaries

- Reduction of cost and complexity; material cost reduction through the elimination of expensive proprietary infrastructure
- Full automation of transactional processes — from payment through settlement

Blockchain is not recommended for:

- High performance (millisecond) transactions
- Small organisations (no business network)
- Database replacement
- A messaging solution
- Transaction processing replacement

Table 2 lists the features of various open source blockchain products with respect to critical blockchain characteristics.

Summing up, blockchain provides better security while carrying out transactions of any value. This unique and universal technology helps to streamline and automate nearly all customer services and legal contracts, while increasing the transparency and effectiveness of enterprises. However, a lot still needs to be done to apply blockchain technology across various business units. **END** 

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